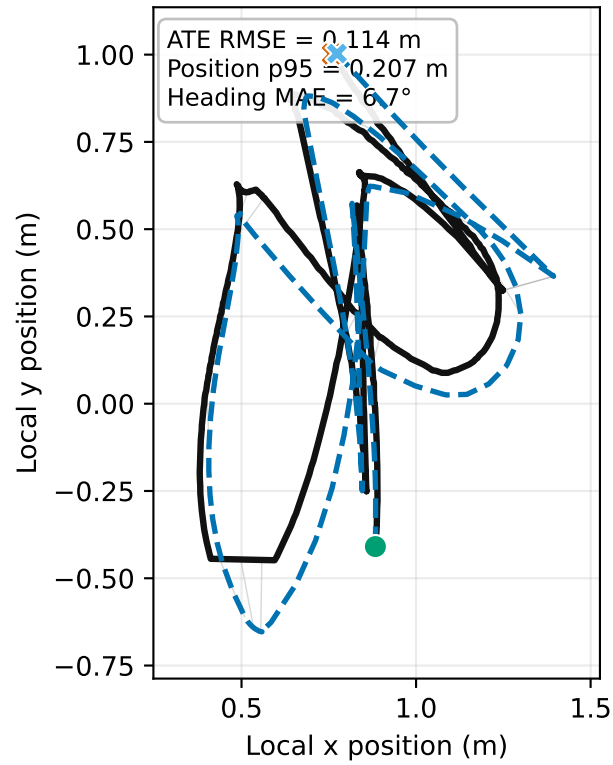


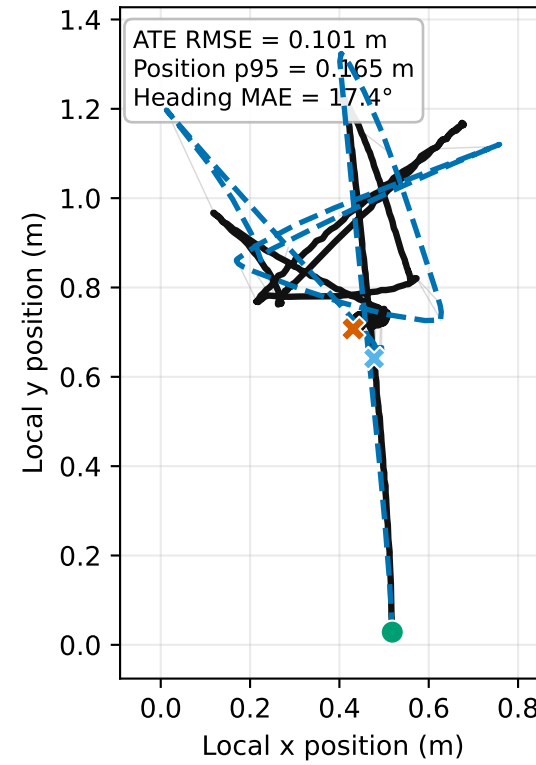
UGV01 Asset-Specific Digital-Twin Instantiation

— Physical UGV01 (AprilTag) - - - Asset-specific digital twin ● Start ✕ Physical end ✕ Twin end

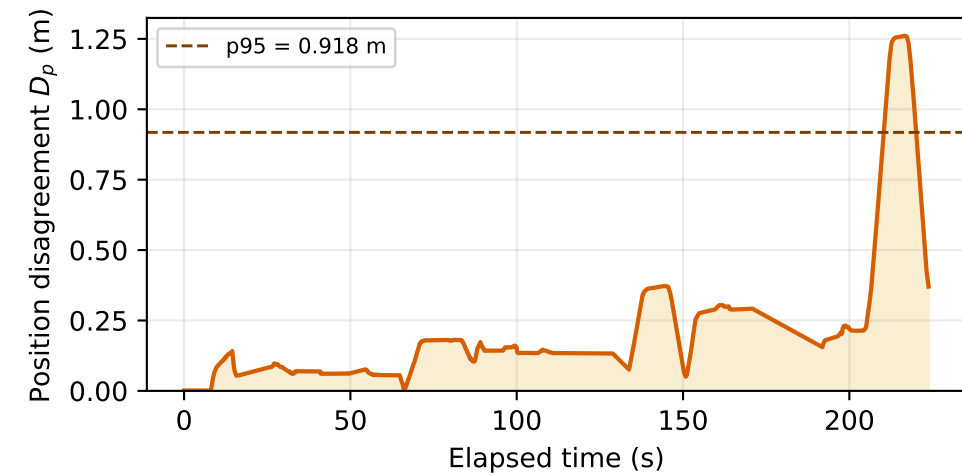
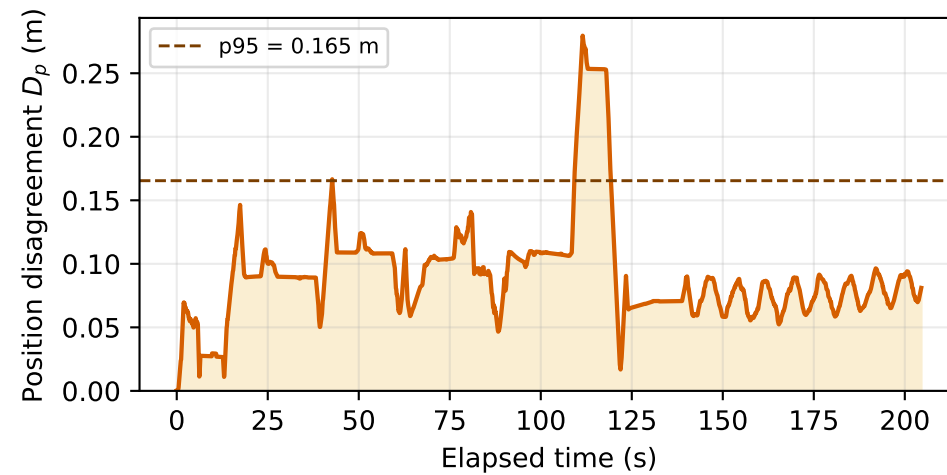
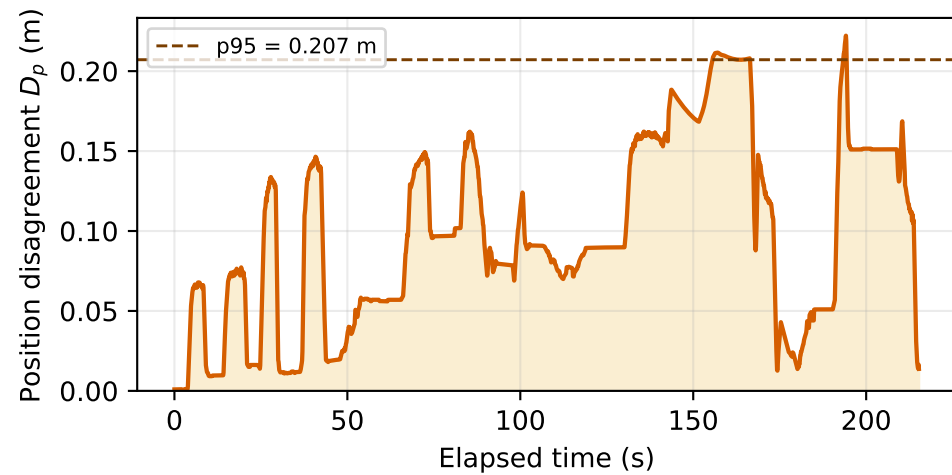
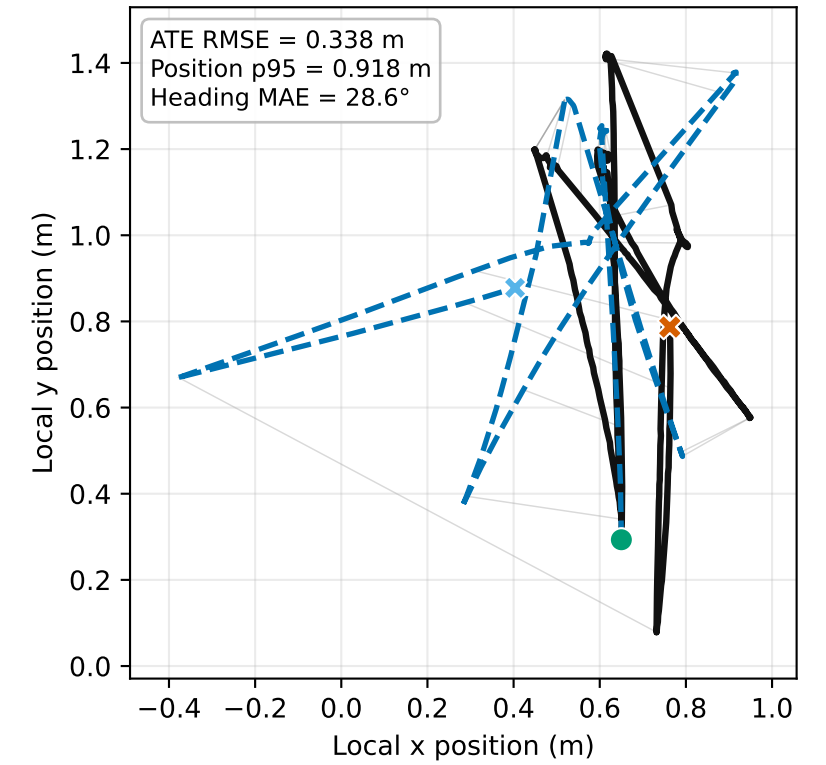
Low-speed carpet



Smooth-floor trapezoid



Smooth-floor 1.5 m square



Solid black: independently observed physical trajectory. Dashed blue: instantiated digital-twin trajectory. Thin connectors show synchronized physical-virtual disagreement.